CCEN 356 Computer Networks

Course Project #1: Design, Implementation, and Analysis of RIP-Based Dynamic Routing Using Cisco Routers and Python Automation in a school network

Project Overview:

This project introduces students to dynamic routing focusing on the Routing Information Protocol (RIP), one of the foundational distance vector routing protocols used in IP network. Students will design a multi-router topology using Cisco 2901 routers, configure RIP, analyze routing behavior, and build Python scripts to automate configuration tasks and collecting routing data.

The project blends hands-on networking with software-based automation, reflecting real-world network engineering workflows.

Learning Objectives:

1. Networking Skills

a. Understand the principles of distance vector routing and RIP operation.

b. Configure RIP on Cisco 2901 routers using Cisco IOS CLI.

c. Build and troubleshoot multi-router topologies.

d. Analyze routing tables, convergence behavior, and route updates.

2. Programming Skills

a. Use Python to automate router configuration via:

- Telnet/SSH (using paramiko or netmiko)
- Parsing routing tables and logs

b. Visualize routing behavior

Project Requirements:

Design the network infrastructure for the following school:

There are three main routers connected to each other; one in the library, one in the teacher's room, and one in the laboratory. Each room has a switch connecting all the devices. For each room, assign several devices, such as laptops, PCs, and printers.

A. Network Topology

- 3 Cisco 2901 routers
- 3 Cisco Switches
- At least two redundant paths between networks
- Ethernet links and Serial Links (where applicable)

B. RIP Configuration Tasks

- Enable RIP version 2
- Configure appropriate network statements
- Disable auto-summary
- Verify routing tables (show ip route)
- Capture RIP updates using:
    - o Debug ip rip
    - o Python scripts

C. Python Automation Tasks

Students will write Python scripts to:

- Connect to routers using SSH (recommended library: netmiko)
- Push configuration commands (e.g., enabling RIP, adding networks)
- Retrieve routing tables and store them in either JSON, CSV, or a simple text log
- Analyse routing changes after link failures or topology changes.

Project Deliverables:

A. Written Report

- Network topology diagram
- RIP configuration details
- Python automation scripts
- Routing table snapshots
- Analysis of:
    - o Convergence time
    - o Route changes after link failures

- o Limitations of RIP

B. Demonstration

- Router connectivity
- RIP operation
- Python Automation in action

C. Source Code Repository

- Python scripts
- Configuration files (.txt exports from routers)
- Documentation